

University Ratings

BAIS:2800

Introduction

When people think about what makes a university “good”, they think of professors, research, student life or reputation. However, there are many different factors that determine a university’s overall success. A university may have relatively low student engagement but may have high ratings on factors such as teaching or research. Universities can be successful in different aspects, but it is important to look further than just their overall rating.

Our dataset looks at world universities from the 2025 Times Higher Education World University Rankings and compares factors such as teaching, research, commercial impact, engagement, and more, to see how universities differ and to determine what patterns we can find. We use this information to compare universities and regions and look for patterns between these different factors and overall ratings. This analysis could be useful in understanding what areas are most important in evaluating colleges. It can also show us how universities differ based on regions, certain areas, and overall strengths. We hope our analysis will create a stronger understanding of what factors are related to university ratings and what other factors may contribute to differences in rankings between universities.

Problem Statement

Plain English: What institutional factors impact a university’s overall world classification or ranking success?

Analytics Framing: In analytics terms, we will be using multiple regression analysis to find relationships between certain characteristics and a university’s overall ranking. Based on 2025 Times Higher Education World University Rankings, we will consider specific factors such as teaching quality, research quality, citation impact, commercial impact, global engagement, student enrollment, faculty ratio, international student percentage, geographic region, and overall university rating. Below is a detailed description of the variables used:

- Rating (Dependent): Overall university rating (0-100)
- Teaching (Independent): Rating of teaching quality (0-10)
- Research (Independent): Rating of research quality (0-10)
- Region (Independent): Continent where university is located (Africa, Asia, Europe, North America, Oceania, South America)
- Citations (Independent): Rating of citation volume for research published by university faculty (0-10)

- Commercial Impact (Independent): Rating of amount of research income earned from industry (0-10)
- Engagement (Independent): Rating global engagement based on student/staff diversity and international research collaborations (0-10)
- Students (Independent): Total students
- Faculty Ratio (Independent): Student-faculty ratio (higher values=larger class sizes)
- Coeducational (Independent): University admits men and women (Yes, No)
- International (Independent): Percentage of international students

Data

This project uses data about university ratings from the 2025 Times Higher Education, which contains information on universities around the world. This data includes information about the universities' region, quality of teaching rating, research quality rating, the amount of students, student to faculty ratio, whether the university is coeducational, the percentage of international students, the overall university rating, and more.

The original dataset contains 1,904 rows and 13 columns. The variables included in the dataset are ID, University, Region, Teaching, Research, Citations, Commercial Impact, Engagement, Students, Faculty Ratio, Coeducational, International, and Rating. We made the following changes to the raw data:

- Selected a random sample of 200 universities out of the 1,904 rows, using the RAND() function in Excel.
- Excluded from the ID and University columns from the analysis because, as identifying variables, they do not provide information about factors related to the ratings of the universities.

The dataset used for our analysis is submitted as “UniversityRatings.xlsx”

One limitation of the data is that it only represents the universities that were included in the 2025 Times Higher Education World University Rankings, meaning these results may not represent every university worldwide. The universities are also located in different regions and may operate in different educational and cultural environments, which may influence the overall ratings. Our analysis focuses on the variables provided in the dataset, which means there may be other factors that influence university ratings but were not included in our model. Because our analysis uses data from 2025 rankings, the relationships we find may not remain the same in the future. Despite these limitations, the variables included in the dataset allow us to examine relationships between characteristics and overall university ratings.